Protein Structure Prediction

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Introduction

AlphaFold 2 (2021) achieved near-experimental accuracy in predicting protein 3-D structure from amino-acid sequence, transforming structural biology. RoseTTAFold and ESMFold offer competitive alternatives with different speed/accuracy trade-offs. Downstream analyses include structural alignment, docking, and mutation impact prediction.

Prerequisites

Amino-acid sequence; familiarity with protein structure (secondary/tertiary elements).

Theory

AlphaFold uses attention-based deep learning trained on PDB structures and multiple-sequence-alignment (MSA) features. The model outputs per-residue (x, y, z) coordinates plus a per-residue confidence score (**pLDDT**, 0-100) and an inter-residue confidence (**PAE**).

pLDDT > 70 = confident; 50-70 = low confidence; < 50 = disordered/unreliable.

Assumptions

The protein has evolutionary neighbours (sufficient MSA) or is not unusually divergent; post-translational modifications and co-factors are not modelled.

R Implementation

AlphaFold prediction itself runs outside R (ColabFold, AlphaFold Docker, OpenFold). R is used for structural analysis via `bio3d`:

```r
library(bio3d)

# Read a PDB (here a built-in example: human CDK2)
pdb <- read.pdb(system.file("examples/hivp.pdb", package = "bio3d"))
summary(pdb)

# Secondary structure, residue count, chains
head(pdb$atom[, c("resno", "resid", "chain", "elety")])

# Torsion angles
tor <- torsion.pdb(pdb)
head(tor$phi)

# Structural alignment of two chains (illustrative)
chain_a <- atom.select(pdb, chain = "A", elety = "CA")
chain_b <- atom.select(pdb, chain = "B", elety = "CA")
xyz_a <- pdb$xyz[chain_a$xyz]
```

```
xyz_b <- pdb$xyz[chain_b$xyz]
c(n_a = length(xyz_a) / 3, n_b = length(xyz_b) / 3)
```

Output & Results

A parsed PDB structure with atoms, residues, and chains accessible programmatically; torsion angles (phi, psi) define the backbone conformation.

Interpretation

"AlphaFold prediction of the novel kinase scored pLDDT 88.3 overall; the activation loop (residues 185-210) had pLDDT < 60, consistent with conformational flexibility seen in homologous kinases."

Practical Tips

- Always check pLDDT per residue; low-confidence regions are unreliable for structural analysis.
- PAE plots reveal inter-domain uncertainty; two confident domains with high PAE between them means the relative orientation is unreliable.
- For mutation impact, combine AlphaFold + molecular-dynamics simulation with Gromacs or AMBER.
- AlphaFold-Multimer handles complexes; standard AlphaFold is monomer-focused.
- ESMFold is faster but less accurate for novel folds; use for screening large sequence sets.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat